

Lab Report

Steady Circle

Participants:

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Group-Number: 5

Date of Experiment: 16th April 2026

Date of Submission: 18th May 2026

<i>Experiment</i>	<i>Date of Experiment</i>	<i>Date of Submission</i>
ACC		
CISS		
HIL	9th April 2026	18th May 2026
RADAR		
SENSOR&ACTUATOR		
STEADY CIRCLE	16th April 2026	24th April 2026

1. Introduction

The aim of this specific lab was to analyze how the influence of various vehicle parameters on lateral dynamics with the help of CarMaker and then use those data files for evaluating graphs on MATLAB. The test setup is a steady-state driving maneuver, allowing the assessment of the vehicle responding to continuous cornering. The goal was to understand how the vehicle responded to variations like *center of gravity*, *stabilizer stiffness*, and *tire characteristics*.

2. Experiments

Simulation in CarMaker

Test Series - Matrix				Variations															
Nr.	Parameter	Designation	Unit	Var1.	Var2.	Var3.	Var4.	Var5.	Var6.	Var7.	Var8.	Var9.	Var10.	Var11.	Var12.	RT_165_70R13	RT_225_60R16		
1	Pos. Center of body	x_Bdy	m	2.241	1	1.5	2	2.5	3	2.241									
2_F	Stabiliser	A_stabiF	N/m	1 (= 15000)						5	10	1	1	5	10	1 (= 15000)			
2_R		A_stabiR								1	1	5	10	5	10				
3	Type of Tyre	RT	-	RT_295_30R18												RT_165_70R13	RT_225_60R16		

In total, 12 variations were used, plus an additional 2 configurations for the tires using the Test Manager on CarMaker. We then used IPG Control to plot parameters Lateral Acceleration and Steering Angle.

Processing in Matlab

We then used MATLAB to use the files obtained from CarMaker for postprocessing. Using the files we were able to simulate the behavior of the vehicle based on the different variations. Through the plots generated by our MATLAB script, we were able to come to an evaluation for each of our mentioned evaluation tasks.

```
path = 'C:\CM_Projects\PVE-VCN_SteadyCircle_Student_11_Ok\SimOutput\20260416\';
Var1 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152419x_Bdy
1.erg']);
Var2 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152505x_Bdy
2.erg']);
Var3 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152526x_Bdy
3.erg']);
Var4 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152542x_Bdy
4.erg']);
Var5 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152605x_Bdy
5.erg']);
Var6 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152620x_Bdy
6.erg']);
Var7 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152701Stabi 5-
1.erg']);
Var8 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152719Stabi 10-
1.erg']);
```

```

Var9 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152735Stabi 1-5.erg']);
Var10 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152755Stabi 1-10.erg']);
Var11 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152811Stabi 5-5.erg']);
Var12 = cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_152825Stabi 10-10.erg']);
Var13 =
cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_RT_165_70R13_New_152911.erg']);
Var14 =
cmread([path,'HAW_Praktika_HAW_Praktika_SteadyCircle_R8_RT_225_60R16_New_163313.erg']);
Legend_1_6 = {'Bdy 2.241', 'Bdy 1', 'Bdy 2', 'Bdy 3', 'Bdy 4', 'Bdy 5'};
figure(1)
hold on
plot(Var1.Car_v.data*3.6,(Var1.Car_YawRate.data./Var1.DM_Steer_Ang.data))
plot(Var2.Car_v.data*3.6,(Var2.Car_YawRate.data./Var2.DM_Steer_Ang.data))
plot(Var3.Car_v.data*3.6,(Var3.Car_YawRate.data./Var3.DM_Steer_Ang.data))
plot(Var4.Car_v.data*3.6,(Var4.Car_YawRate.data./Var4.DM_Steer_Ang.data))
plot(Var5.Car_v.data*3.6,(Var5.Car_YawRate.data./Var5.DM_Steer_Ang.data))
plot(Var6.Car_v.data*3.6,(Var6.Car_YawRate.data./Var6.DM_Steer_Ang.data))
hold off
legend(Legend_1_6)
ylim([-0.2,0.8])
xlim([30 120])
xlabel("speed [km/h]"); ylabel("yaw gain")
figure(2)
hold on
plot(Var1.Car_ay.data, rad2deg(Var1.DM_Steer_Ang.data))
plot(Var2.Car_ay.data, rad2deg(Var2.DM_Steer_Ang.data))
plot(Var3.Car_ay.data, rad2deg(Var3.DM_Steer_Ang.data))
plot(Var4.Car_ay.data, rad2deg(Var4.DM_Steer_Ang.data))
plot(Var5.Car_ay.data, rad2deg(Var5.DM_Steer_Ang.data))
plot(Var6.Car_ay.data, rad2deg(Var6.DM_Steer_Ang.data))
hold off
legend(Legend_1_6)
xlabel("lateral acceleration [m/s^2]"); ylabel("steering angle [°]")
ylim([0,60]);xlim([0 9])

```

3. Evaluation tasks

COG

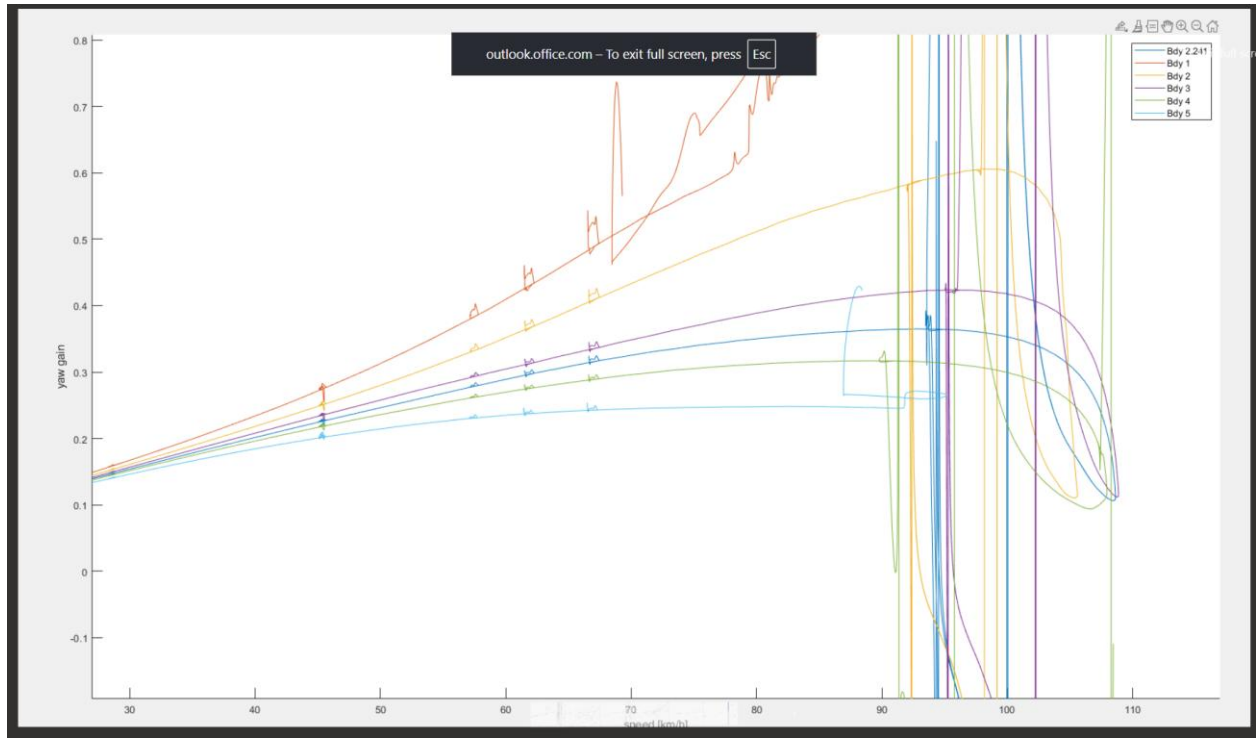


Fig. 1: Yaw Gain (yaw rate / steering angle) vs. Vehicle Speed — Center of Gravity Variants (Bdy 1–5 and original Bdy 2.241)

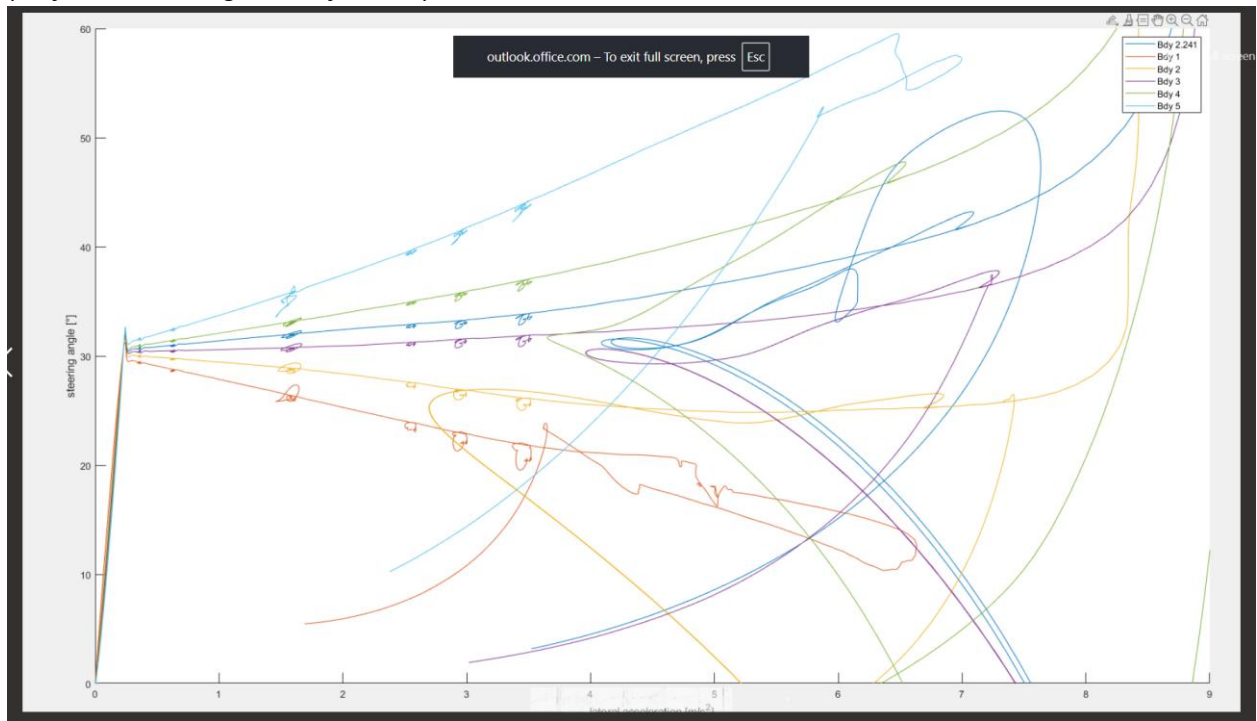


Fig. 2: Steering Wheel Angle vs. Vehicle Lateral Acceleration — Center of Gravity Variants (Bdy 1–5 and original Bdy 2.241)

Steering Behavior Classification

The changes in center of gravity positions (x_{bdy}) are listed below in a table format where each variant value is named.

Variant	x_{Bdy} (m)	Character
Bdy 1	1.0	oversteering
Bdy 2	1.5	weakly oversteering, neutral
Bdy 3	2	mildly understeering
Bdy 2.241 (original)	2.241	understeering
Bdy 4	2.5	understeering
Bdy 5	3	Strongly understeering

Justification:

Moving the CoG rearward (small x_{Bdy}) shifts vertical load toward the rear axle. The rear tires saturate earlier than the front, so the rear slip angle exceeds the front's, causing oversteer. Conversely, a CoG closer to the front (large x_{Bdy}) overloads the front tires; they saturate first, resulting in an understeer. Bdy 1 sits at the rearward extreme, and Bdy 5 at the forward extreme, giving the monotonic progression seen above.

Characteristic/Critical Velocities

Variant	v_{ch} (understeer) / v_{crit} (oversteer) in km/h
Bdy 1	88 (v_{crit} — oversteering)
Bdy 2	100
Bdy 3	99
Bdy 2.241	93
Bdy 4	93
Bdy 5	87

Trend:

As x_{Bdy} grows from 1.0 to 3.0 m, the vehicle moves from oversteer through near-neutral into progressively stronger understeer; v_{ch} falls from ~100 to ~87 km/h, and peak heights drop in lockstep, reflecting the steering gradient growing as the front axle is loaded more.

Relation between v_{ch} and COG Position (l_v, l_h)

- v_{ch} decreases as l_h increases (CoG closer to front axle).
- Equivalently, v_{ch} increases with l_v (CoG closer to rear, within the understeering regime)

Physical interpretation: a larger l_h loads the front axle more heavily, so the front tires saturate earlier, the self-steering gradient grows, and the yaw-gain peak (which by definition lies at v_{ch}) shifts to a lower speed. Once l_h is small enough that $l_h \cdot c_{\alpha v} < l_v \cdot c_{\alpha h}$, steering gradient turns negative— v_{ch} no longer exists, and v_{crit} becomes the relevant indicator

Impact of Stabilizer:

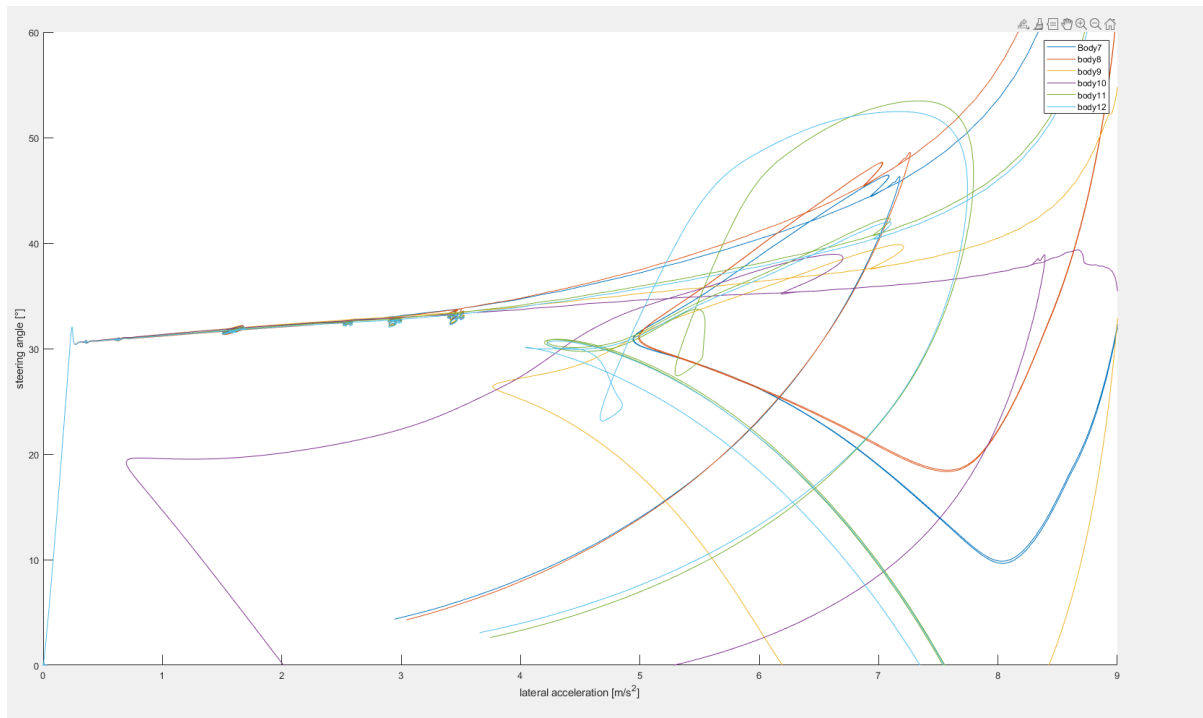


Fig. 3: Yaw Gain (yaw rate / steering angle) vs. Vehicle Speed — Stabilizer Stiffness Variants (Bdy 7–12)

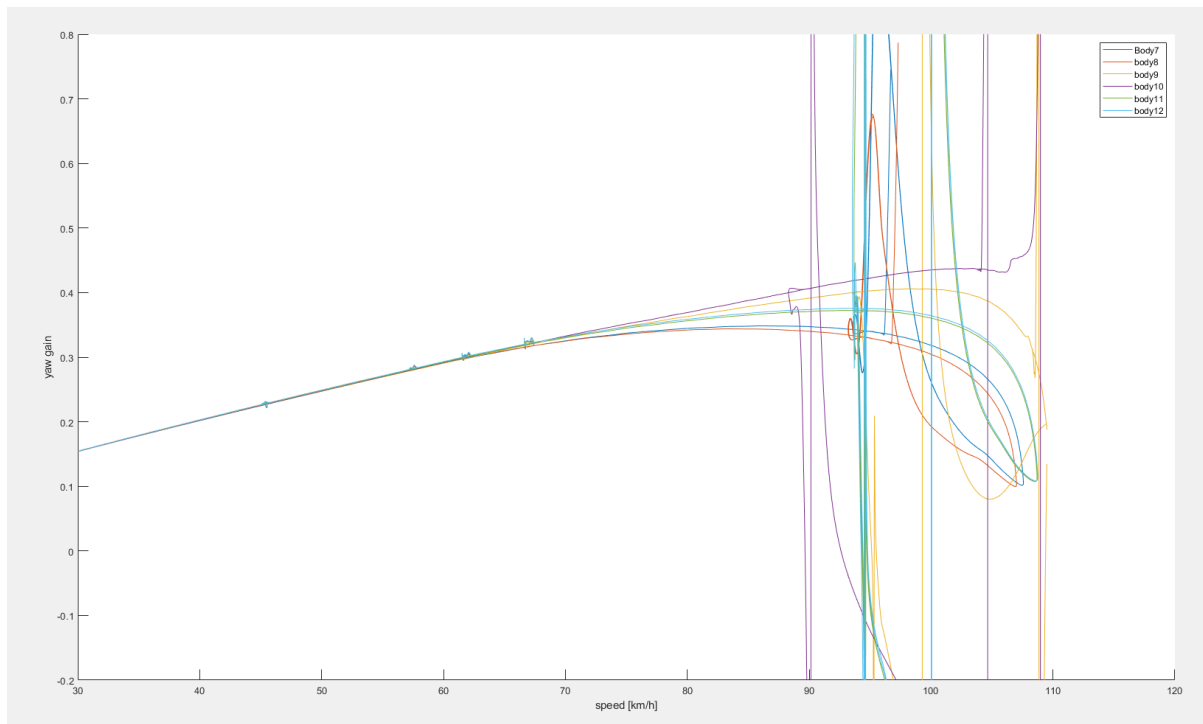


Fig. 4: Steering Wheel Angle vs. Vehicle Lateral Acceleration — Stabilizer Stiffness Variants (Bdy 7–12)

Steering behavior

Body	<u>trend</u>	<u>characteristic</u>
Bdy 7	positive slope, smooth finite peak (~ 0.35)	understeering
Bdy 8	weakly positive, very tall, narrow peak (~ 0.67)	weakly understeering, near-neutral
Bdy 9	positive, smooth slope peak (~ 0.40)	understeering
Bdy 10	flat / slightly negative, no finite peak, near-vertical asymptote	oversteering
Bdy 11	positive slope smooth peak (~ 0.37)	understeering
Bdy 12	positive smooth peak (~ 0.37)	understeering

An anti-roll bar increases wheel-load transfer on its axle during cornering. Because tire cornering stiffness decreases with vertical load, the axle with the stiffer stabilizer saturates first.

Therefore:

Stiff front stabilizer $\rightarrow$ understeer.

Stiff rear stabilizer $\rightarrow$ oversteer.

Balanced stiffening of both axles $\rightarrow$ load transfer rises on both sides,

- The vehicle showing oversteer (body 10) therefore corresponds to the configuration with a strongly stiffened rear bar relative to the front; the near-neutral body 8 corresponds to a setup where the rear stabilizer is the stiffer of the two but less extremely so; and the four understeering bodies correspond to either front-biased stiffening or balanced stiffening.

Characteristic Speed Summary:

Stiffening only the front axle (body 7) or stiffening both axles together (body 11, body 12) produces moderate understeer with v_{ch} in the 90–95 km/h range and peaks around 0.35–0.37. A strongly front-biased setup (body 9) increases v_{ch} to ~ 100 km/h with a slightly higher peak. Rear-biased stiffening drives the vehicle toward and across the neutral boundary: body 8 sits just inside understeer with a very tall peak (~ 0.67), while body 10 crosses into oversteer entirely, with no v_{ch} but a critical velocity of ~ 88 –90 km/h.

Impact of Tire

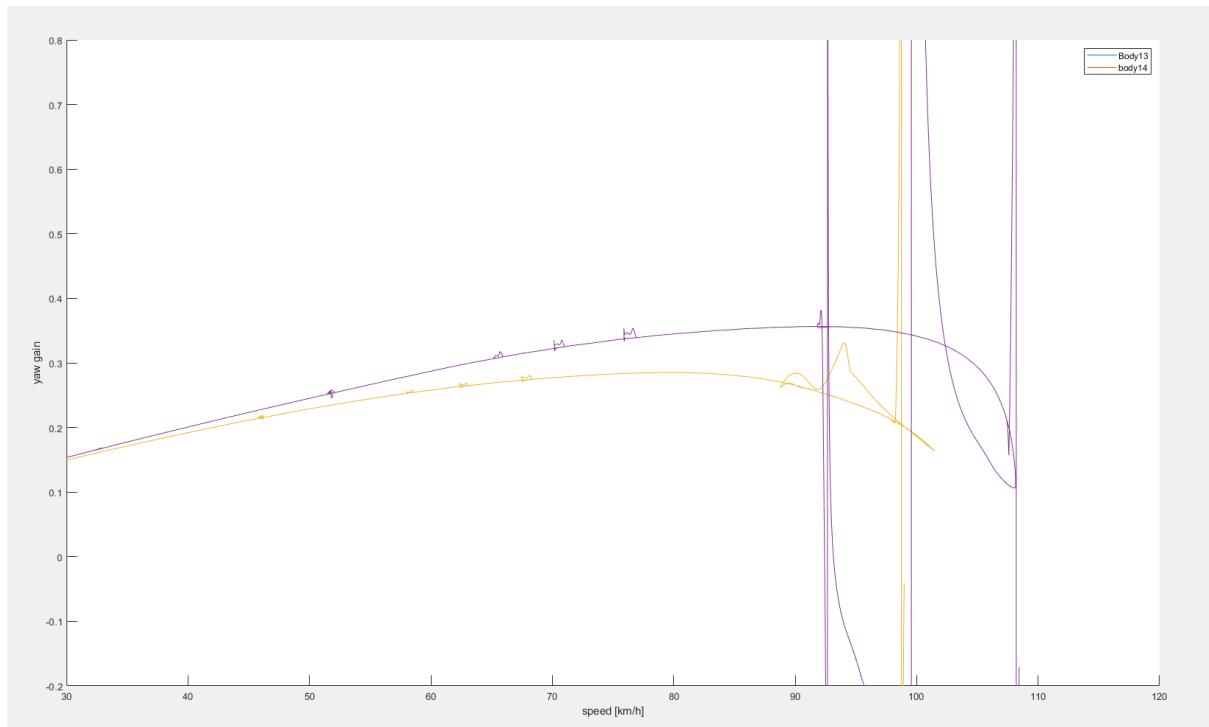


Fig. 5: Yaw Gain (yaw rate / steering angle) vs. Vehicle Speed — Tyre Variants (Bdy 13: RT 165/70R13, Bdy 14: RT 225/60R16)

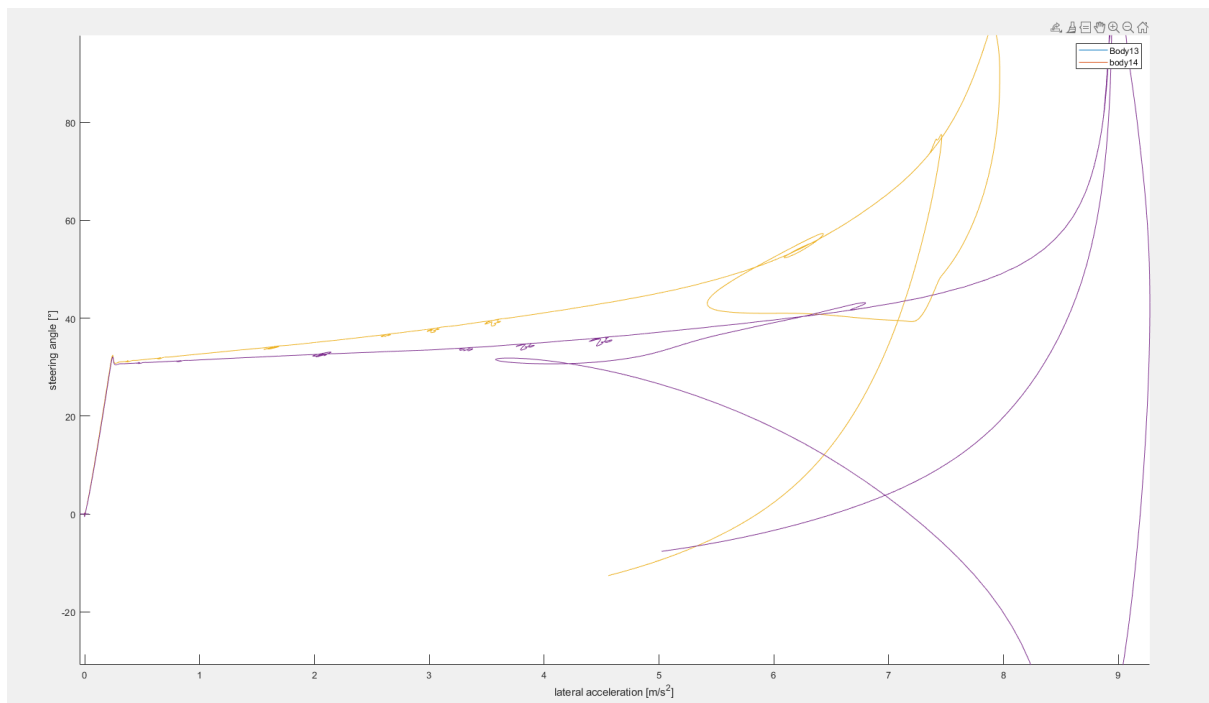


Fig. 6: Steering Wheel Angle vs. Vehicle Lateral Acceleration — Tyre Variants (Bdy 13: RT 165/70R13, Bdy 14: RT 225/60R16)

Characteristic/Critical speed

<u>Body</u>	<u>Type</u>	<u>v_{ch}</u>
Bdy 13	understeer	80 km/h
Bdy 14	understeer	75 km/h

Reasoning:

Below ~50 km/h, both curves overlap → identical linear behavior (tires still in their linear range). Each curve then rises, plateaus, and bends over—the apex of that arch is v_{ch} . Body 14's apex is lower and further left, meaning stronger understeer, meaning a smaller v_{ch} . Body 13's apex is higher and further right, meaning milder understeer causing a larger v_{ch} . The vertical spikes past ~92 km/h are simulation artifacts at the grip limit, not v_{crit} asymptotes—both curves had already turned over before them, confirming understeer rather than oversteer. Overall, the higher and further right the yaw-gain peak sits, the milder the understeer and the larger v_{ch} ; the lower and further left it sits, the stronger the understeer and the smaller v_{ch} .

Both curves start from the same point at low lateral acceleration

- As a_y increases, both curves slope **upward** → both vehicles need *more* steering as cornering load builds → both are understeering.
- The yellow line (body 14) is **noticeably steeper** than the purple line (body 13)—for the same lateral acceleration, body 14 demands a larger steering angle.
- Neither curve slopes downward at any point in the linear region → neither vehicle crosses into oversteer.
- Above ~6 m/s² both curves enter the non-linear regime (loops and sharp rises) where the steady-state interpretation no longer holds — only the slope near the origin is used for classification.

Conclusion:

The steady-state circular test in CarMaker provided a clear, hands-on way to characterize the lateral dynamics of a vehicle by driving it on a constant-radius circle with slowly increasing speed and varying three parameters: center-of-gravity position, stabilizer stiffness, and tire properties. From the resulting $\delta(a_y)$ and $\psi/\delta(v)$ diagrams, each configuration could be classified as understeering, neutral, or oversteering, and the corresponding characteristic or critical speed could be extracted.

The evaluation showed that mass-distribution changes—shifting the CoG or unbalancing the stabilizers—were able to flip the steering character entirely between oversteer, neutral, and understeer, whereas tire changes only modulated the strength of understeer without crossing the neutral boundary. This confirmed that the steering character is fundamentally governed by which axle saturates first and that this is far more sensitive to load distribution than to absolute tire grip. The steady-state circular test therefore proved to be an effective and reproducible method for quantifying these effects and for guiding the tuning of a vehicle toward predictable, safe cornering behavior.

